

Independent AI Evaluation Project

Environmental reasoning in a protected coastal landscape: stone, evidence and inference

PREPARED: 27 September 2026

MODELS EVALUATED: Microsoft Copilot Auto / Google Gemini 3.1 Pro

PROJECT TYPE: Independent, source-grounded AI evaluation

CASE: A small, dry-laid loose-stone land-art installation in a protected coastal landscape in southern Spain

Public portfolio case study. Location, institutional and source identifiers have been generalised throughout to protect the confidentiality of the underlying project. Raw model outputs, scoring sheets and primary case records are retained separately in the audit archive.

Abstract

This project evaluates how two general-purpose AI systems reason from incomplete, multi-layered environmental evidence, using a real land-art case study set in a protected coastal and geological landscape. The benchmark was built before scoring from a controlled source pack combining a case-specific authority communication, project records, official geological and protected-area sources, regional coastal and park-management context, and two peer-reviewed analogue studies. Ten closed-source prompts tested geology, causal reasoning, biodiversity scale, institutional precision, restoration logic, visitor pressure and environmental communication. Two additional open-web stretch prompts tested current-source research and operational due diligence.

Responses were saved unedited, anonymised as X/Y, scored blindly on a weighted 100-point rubric, and attributed only after scores were locked. Copilot Auto scored 1,156/1,200 (96.3%); Gemini 3.1 Pro scored 945/1,200 (78.8%). The central finding was not simply a difference in factual recall. The larger distinction was evidential discipline: preserving spatial scale, source status, modal language, causal uncertainty and administrative nuance. Gemini remained strong on several controlled prompts but overreached more often on integrative tasks and refused the final stretch prompt entirely.

Portfolio capabilities demonstrated

This case study demonstrates an end-to-end AI-evaluation workflow: source selection and hierarchy design; pre-registered ground truth and scoring benchmarks; controlled prompt construction; blind comparative adjudication; hallucination and overclaim detection; causal and uncertainty analysis; live-web citation auditing; reproducible score aggregation; and preservation of raw outputs, metadata and audit evidence.

1. Case background

The project began several years ago with a small classical labyrinth, documented as approximately three metres in size, laid by hand in loose stone without mortar, cutting or excavation. That construction description matters because it reduces some obvious forms of physical intervention, but it does not resolve the environmental question. A central rule of this evaluation was therefore established before any model output was seen: dry-laid construction does not prove zero impact, while unauthorised status does not itself prove significant ecological damage.

The environmental setting is unusually suitable for testing AI reasoning because several evidence layers overlap. Official regional-government material describes the local rock formation as a Late Pleistocene fossil dune system formed from wind-transported carbonate grains, including ooids, with preserved cross-bedding. National geological-heritage records list the site as a georesource of principal sedimentological interest. Wider protected-area material adds EU-level habitat-network context, but that evidence applies at site or park scale and cannot be used to place specific species or habitats at the exact installation footprint without local survey evidence.

The strongest case-specific evidence is a written communication from the relevant protected-area authority to the project, sent in September 2026 (S0). The authority states that the installation lacked administrative

authorisation, was on state-owned coastal public domain land, and lay within a delimited geological-heritage zone. It says that artificial placement and arrangement of stones modifies the space and "may affect" the values that motivated the geological inventory. That wording is important: "may affect" expresses possibility or risk, not a quantified measurement of ecological damage.

S0 also states that promotion of the installation as a freely accessible resource was favouring visitor and group movement off established routes, with effects on soil, vegetation, geological values and fauna. The authority says it will remove the installation and its elements to restore the space and avoid consolidation of unordered public use. It also conditionally warns that reinstating this type of structure or activity could, where applicable, lead to a sanctioning proceeding. The communication does not itself notify an already-open sanctioning proceeding, fine, deadline or appeal route.

A separate communication issue also matters. The authority states that the project is not part of the wider geopark's network of collaborating entities and that no authorisation or agreement permits the installation to be presented as carried out in collaboration with, promoted by or backed by the geopark or park administration.

These distinctions made the case appropriate for AI evaluation. A model could fail by minimising the intervention because it was small and dry-laid, but it could also fail by converting an administrative and management dispute into scientific proof of severe ecological harm. The project was designed to reward neither direction. It rewarded traceable evidence, scale discipline and calibrated judgment.

2. Evidence base and source hierarchy

The evidence pack used four classes:

- **A — Primary case evidence:** the original authority communication, dated project records, and archived/public project material.
- **B — Official/statutory evidence:** national and regional geological, environmental and statutory sources.
- **C — Peer-reviewed scientific evidence:** external studies used for mechanisms and analogues.
- **E — Reasoned inference:** conclusions that follow from the evidence but are not direct observations.

The source hierarchy was deliberately strict because environmental answers can sound persuasive while mixing evidence types that do not support the same proposition.

2.1 Geological and official sources

Regional-government geological trail material supports the Late Pleistocene origin of the fossil dunes, the role of ooids and wind transport, cross-bedding and present coastal/alluvial processes. National geological-heritage records support geological-interest inventory status and sedimentological significance. Habitat-network material provides EU-level protected-area context at the wider site scale. Regional park-management legislation provides the protected-area management framework. International geopark recognition provides geopark context only. National coastal law provides the general legal framework for uses of state-owned coastal public domain land.

A key evaluation rule followed from this structure: these designations are not interchangeable. Park management status, EU habitat-network status, international geopark recognition, geological-interest inventory status and coastal public-domain law are different legal or management layers. None can be treated as proof that another body authorised or endorsed the installation.

2.2 Scientific analogue evidence

S8, Sanromualdo-Collado et al. (2023), examined persistent stone wind shelters in an active dune system on a different part of the Spanish coast. The study reported changes in airflow, sediment transport, sediment accumulation and vegetation. This is scientifically relevant because it demonstrates plausible physical mechanisms around stone structures in an arid coastal dune setting. It is not direct evidence that a three-metre loose-stone labyrinth beside fossil eolianites at this specific site produced equivalent effects.

S9, Rocha et al. (2020), discusses stone stacking as a potential threat to rock-dwelling biodiversity and notes that surface rocks can function as attachment, refuge and foraging microhabitats. Again, the study supplies mechanism and context. It does not establish which species or habitats occurred at the installation's footprint or

what magnitude of impact occurred there.

The evaluation therefore penalised "citation laundering": using a legitimate scientific source to support a proposition it did not actually establish at the case site.

3. Evaluation method

The benchmark was fixed before the main model outputs were scored. This was important because it prevented the marking standard from drifting in response to persuasive answers.

3.1 Models and run conditions

The main comparison used:

- Google Gemini 3.1 Pro, standard/default mode, Extended Thinking off.
- Microsoft Copilot Auto.
- Browser: Edge.
- Chat state: fresh chat for each controlled run.
- Prompts 1–10: browsing/search prohibited; models could use only the supplied source packet.
- Prompts 11–12: open-web stretch round; live research was allowed and required.

Earlier Copilot Smart pilot runs were preserved but excluded from the main comparison because the live interface available for the controlled series exposed Auto rather than Smart. An earlier Prompt 3 run was also invalidated after a browser-session/context disruption. Those outputs remain audit evidence but are not included in the paired score totals.

3.2 Blind scoring

For every prompt, both raw answers were stored before scoring. They were labelled RESPONSE X and RESPONSE Y without model identity. A 100-point rubric was then applied:

Criterion	Weight
Factual accuracy	25
Evidence / citation discipline	15
Reasoning quality	15
Uncertainty calibration	10
Local-vs-general distinction	10
Management / legal precision	10
Environmental nuance	10
Writing / instruction following	5

Deductions were written against specific source or benchmark failures. Only after the score file was locked was model identity revealed and appended. Raw outputs, metadata, score sheets and manifest updates were mirrored between the local project store and a cloud backup and hash-checked.

3.3 Prompt design

The twelve prompts increased in integrative difficulty.

Prompts 1–4 tested geological explanation, the claim that dry-laid construction meant "no impact," transferability of the analogue study, and plausible geoheritage impact pathways. Prompts 5–8 tested biodiversity scale, protection layers, removal/restoration documentation and a non-intrusive visitor-pressure research design. Prompts 9–10 tested balanced art/environment critique and the environmental trade-offs of moving from physical intervention toward book/browser-based experience. Prompts 11–12 tested open-web source auditing and permission-first future practice.

The design deliberately rewarded answers that said "the available evidence does not establish that" when that was the correct conclusion.

4. Results

4.1 Prompt-level scores

Prompt	Test focus	Copilot Auto	Gemini 3.1 Pro
P01	Geological explanation	99	98
P02	"Dry-laid means harmless?"	99	97
P03	Analogue-site transferability	100	96
P04	Geoheritage impact pathways	96	86
P05	Biodiversity at exact footprint	100	93
P06	Protection layers	100	90
P07	Removal/restoration reasoning	97	92
P08	Visitor-pressure research design	99	88
P09	Art/environment critique	93	77
P10	Physical to print/digital continuation	97	80
P11	Open-web public-copy audit	90	48
P12	Open-web future due diligence	86	0
Total		1,156 / 1,200	945 / 1,200
Mean		96.3	78.8

For the controlled closed-source round alone, Copilot scored 980/1,000 (98.0) and Gemini scored 897/1,000 (89.7). In the open-web stretch round, Copilot scored 176/200 (88.0) and Gemini scored 48/200 (24.0).

The stretch result needs separate interpretation. Gemini's Prompt 12 output was a complete refusal rather than a substantive environmental answer. That run outcome is relevant to task reliability, but it should not be read as evidence that Gemini's ordinary controlled environmental reasoning was equivalent to a zero-score response. The controlled-round comparison is therefore the cleaner measure of substantive reasoning quality; the stretch round additionally tests research execution and instruction following.

4.2 Criterion-level scores

Across all twelve prompts:

Criterion	Maximum	Copilot	Gemini
Factual accuracy	300	288 (96.0%)	232 (77.3%)
Evidence / citation discipline	180	165 (91.7%)	128 (71.1%)
Reasoning quality	180	176 (97.8%)	147 (81.7%)
Uncertainty calibration	120	119 (99.2%)	89 (74.2%)
Local-vs-general distinction	120	120 (100.0%)	103 (85.8%)
Management / legal precision	120	113 (94.2%)	95 (79.2%)
Environmental nuance	120	118 (98.3%)	100 (83.3%)
Writing / instruction following	60	57 (95.0%)	51 (85.0%)

The largest substantive differences were in evidence discipline and uncertainty calibration. Copilot's strongest feature was not simply getting more facts right. It more consistently kept evidence in the right category and resisted converting possibility, analogy or institutional context into a stronger claim.

Gemini was often accurate on individual source facts but lost points when several evidence layers had to be held apart simultaneously. That pattern became more visible from Prompt 4 onward and was particularly clear in Prompts 8–11.

5. Representative failure modes

5.1 Spatial-scale leakage

Prompt 5 asked for a biodiversity note about the exact installation footprint while supplying wider habitat-network evidence. Both models understood the central issue, but Gemini stated that the exact installation was "inside" the protected habitat area even though the controlled packet did not explicitly establish that spatial relation. It also broadened the authority's reported effects to "across the area."

This is a common environmental reasoning risk. Protected-area data can be highly relevant context without proving that a specific habitat or species was present at a three-metre footprint. The stronger answer preserves the distinction and states what local survey evidence would be required.

5.2 Converting absence of evidence into affirmative fact

Prompt 6 tested overlapping designations and endorsement discipline. Gemini moved from "the supplied evidence does not state that the habitat network or geopark authorised or endorsed the installation" to affirmative statements that those bodies did not do so.

The difference looks small linguistically but matters methodologically. Absence of evidence in a closed packet is not always evidence of absence. The correct move is to state the source boundary, unless another source directly establishes the negative proposition.

5.3 Hardening modal or attributed authority language

Several Gemini answers strengthened S0.

In Prompt 8, the authority communication became a "regulatory determination" and "official qualitative impact evaluation." In Prompt 9, authority-reported effects and external mechanisms contributed to a final categorical statement that the intervention "inherently" altered the living environment and "directly contradicts" the project ethic. In Prompt 10, S0's statement that promotion was "favouring" off-route movement was strengthened into behaviour "triggered by" promotion.

These shifts matter because environmental and administrative evidence often depends on modality. "May affect," "is favouring," "could lead to" and "the authority states" are not interchangeable with "caused," "proved," "determined" or "will be sanctioned."

5.4 Analogue evidence upgraded into local proof

Prompt 3 was explicitly designed to test transferability of S8. Both models performed well. The harder failure appeared later when analogue mechanisms were integrated with other evidence. In Prompt 9, Gemini generalised stone-rearrangement studies more strongly than the source pack supported and used them as part of a categorical moral/environmental conclusion.

The correct scientific use of S8 and S9 is narrower: they demonstrate mechanisms that make certain impacts plausible. They do not demonstrate the magnitude, presence or causal pathway at the case site without a local bridge.

5.5 Substitution and access assumptions

Prompt 10 asked whether moving the project's continuing public experience from a physical site intervention toward a book and browser-based 3D world had environmental advantages.

Gemini correctly recognised that print and digital formats have non-zero impacts, but it treated several benefits as established: that the change "relieves" site pressure, "removes" geographic barriers, "allows broader public access" and "protects physical land from foot traffic."

The evidence supports these as possibilities, not measured outcomes. A digital experience may substitute for physical visitation, but it may also stimulate interest in the place. Access may broaden, but that depends on cost, distribution, devices, connectivity and usability. No project-specific lifecycle data were available to prove a lower net footprint.

5.6 Open-web source auditing

Prompt 11 exposed a different capability: checking what was actually live on the current project website after the authority correspondence.

The pre-run benchmark found that several pages had already been corrected. In the direct live-page retrieval used for the 27 September benchmark, the installation's own webpages substantially removed earlier visitor-invitation language and added independence/context wording. The main remaining issues were narrower: present-tense physical-existence wording on one page, and a homepage link to a third-party locator listing that still presented the installation as a permanent, freely accessible public site. Because these are live web pages, this finding is a dated snapshot rather than a claim about later versions.

Copilot found the third-party listing and correctly distinguished it from the project's own copy, but missed the secondary page issue and produced a 115-word rather than 120-word replacement. Gemini did not provide live-web citations, did not distinguish page ownership clearly, and stated without evidence that the physical installation was "no longer open or accessible."

This prompt shows why "web-enabled" is not equivalent to "source-audited." The evaluator still has to ask whether the model found the live page, whether the cited source actually contains the wording, whether it is current, and whether third-party material is being attributed correctly.

5.7 Operational workflow and refusal

Prompt 12 asked for a permission-first checklist for future protected-landscape work. Copilot produced a strong gated workflow covering ownership, competent authorities, designation layers, exact-site ecology, natural material sourcing, access, visitor pressure, cumulative impacts, permissions, removal/restoration, monitoring and communications. It also included clear STOP rules for refused, unclear, expired or materially changed permission.

Its major weakness was citation discipline: the answer made external claims about the habitat network and geopark designation without providing the required current official citations. This mattered because Prompt 12 was specifically an open-web source-selection test.

Gemini did not attempt the task and returned a refusal. That refusal is recorded as the raw output and scored 0/100 under the pre-existing rubric.

6. What the comparison suggests about good environmental AI reasoning

The strongest answers across this project shared a small number of repeatable behaviours.

First, they preserved source provenance. A park authority communication can establish what the authority told the project. It is not the same thing as a field ecology study. A scientific paper from a different dune system can demonstrate a mechanism. It is not a measurement at the case site. A project's own web page can establish what the project says it built. It is not independent proof of environmental condition.

Second, they preserved spatial scale. Park-wide habitats and species are not automatically footprint-level facts. A designated georesource is not the same thing as a measured impact point. A three-metre installation can have indirect visitor effects larger than its physical footprint, but that also needs evidence.

Third, they preserved modality. "May affect" is not "did damage"; "could lead to" is not "a sanction was opened"; "the authority states" is not the same as an independently quantified ecological effect.

Fourth, they separated mechanism from attribution. A mechanism may be biologically or geomorphologically plausible even when the available evidence cannot show that it occurred locally.

Fifth, they treated uncertainty as information rather than weakness. In several prompts, a sentence such as "the evidence does not establish this at the exact footprint" was a stronger answer than a more fluent but speculative ecological narrative.

Finally, they treated public communication as part of environmental management. Location information, maps, access wording and third-party directories can influence visitor behaviour. Environmental reasoning therefore extends beyond the object itself to the way a project is promoted and accessed.

7. Implications for future practice

The evaluation suggests several concrete principles for future outdoor land-art work in protected or environmentally sensitive landscapes.

7.1 Permission first, not permission after installation

The strongest lesson from this case is procedural. Artistic intent, reversibility and small scale do not substitute for identifying the land status, competent authority and applicable rules before installation.

For any future outdoor intervention, the proposal should be defined before entering the site: exact location, dimensions, duration, materials, installation method, visitor plan, publicity plan and removal method. Landowner and management responsibilities should then be established, followed by review of every relevant designation or legal layer.

If any required permission is unknown, missing, expired or conditional, the project should remain at STOP status until the issue is resolved in writing.

7.2 Natural material is not automatically impact-free material

The original installation was documented as loose stone and dry-laid by hand. That method avoids mortar, excavation and cutting, but the project should not generalise from this to a rule that "found" or natural material is environmentally neutral.

Future material decisions should document provenance, ownership, permission to collect or move material, quantity, transport, placement and removal. In a protected site, even apparently loose stones may form part of the geological, geomorphological or ecological system.

7.3 Treat visitor behaviour as part of the intervention

The authority's strongest environmental concern in S0 was not limited to the physical three-metre form. It included promotion of the work as a freely accessible resource and the off-route movement that the authority associated with that promotion.

Future environmental due diligence should therefore examine not only what the artwork physically does, but what the communication around it may cause people to do. Maps, coordinates, open-access wording, directory listings, social posts, group visits and photographic hotspots can all create secondary pressure.

This implies a pre-launch communications review. Public copy should match the permission status, route conditions and duration of the work. Third-party listings should also be monitored where they materially affect visitor understanding.

7.4 Design removal and restoration before installation

Removal should not be improvised after a problem occurs. A permission-first project should define before installation who is authorised to remove the work, what triggers removal, how materials will be handled, what condition is being restored, and how post-removal condition will be documented.

A pre-installation baseline is important because without it, later photographs can show change but not necessarily establish cause or recovery. Restoration objectives should also not be treated as proof that restoration succeeded; post-removal evidence is still required.

7.5 Keep communication and institutional relationships precise

The S0 correspondence demonstrates that geographical location within a geopark or protected area must not be turned into a claim of collaboration, endorsement or administrative backing.

Future project copy should distinguish: location within a recognised territory; factual reference to a designation; formal collaboration; permission for a specific activity; and institutional endorsement.

Only the latter three require explicit documentary support. Historical pages may be preserved as archive material, but their status should be clear when circumstances change.

7.6 Digital continuation changes the impact profile rather than eliminating it

Moving a project's continuing public experience toward publishing and a browser-based 3D world can reduce dependence on a continuing physical intervention at a sensitive site and may reduce site pressure if it substitutes for physical visitation.

However, this should not be presented as proof of a lower total environmental footprint. Print production, distribution, hosting, network use and devices all have material and energy costs. No project-specific lifecycle comparison was available in this evaluation.

The defensible claim is narrower: digital and print continuation can avoid the need for continued physical installation at the original site, while shifting rather than eliminating environmental impacts.

8. Limitations

This evaluation has several important limits.

8.1 Small model sample. Only two model configurations were compared. The results should not be generalised to all AI systems or even to all modes of the same products.

8.2 Date-specific behaviour. Model behaviour changes over time. The scores describe the versions and interface conditions used on 27 September 2026. The open-web results in particular depend on what each model could retrieve and how the live web appeared on that date.

8.3 One refusal materially affects the aggregate. Gemini's Prompt 12 refusal produced a zero score and materially lowers its all-prompt mean. The controlled-round average therefore provides a better view of its substantive source-grounded reasoning, while the stretch score additionally reflects execution reliability under open-web instructions.

8.4 No ecological field survey. The project did not conduct an exact-footprint biodiversity, soil, geomorphology or visitor-impact survey. It therefore cannot establish the magnitude of ecological effect caused by the installation.

8.5 Authority evidence is not a substitute for quantified ecology. S0 is highly important case evidence because it records the competent authority's position, including its assessment of visitor effects. It does not provide visitor counts, species-specific analysis, an experimental control or a measured effect size.

8.6 Analogue studies are intentionally limited. S8 and S9 were used to test whether models could reason from mechanisms without over-transferring results. The scientific evidence is relevant but does not convert this case into a measured replication of the analogue sites.

8.7 Scoring remains a human judgment. The rubric, source hierarchy and pre-written benchmarks make the process reproducible, but scoring nuanced language still requires judgment. For that reason, every deduction was recorded with a reason, raw outputs were preserved, and identities were hidden until scores were locked.

8.8 Author's relationship to the case. The evaluator is the project lead for the installation discussed in this case study, and scored this material as a self-audit rather than as an independent third party. The evaluation design (pre-registered rubric, blind labelling, locked scoring before identity reveal) was chosen specifically to control for that conflict of interest, but readers should weigh the results accordingly.

9. Conclusion

This case shows why environmental AI evaluation needs more than fact checking.

A model can know that a coastal landscape is protected, recognise that rocks can form microhabitats, and correctly describe a scientific paper, yet still reason badly if it collapses park-wide data into an exact footprint, turns a possible mechanism into a demonstrated local impact, changes an authority's "may affect" into "did damage," or treats a designation as project endorsement.

The strongest performance in this study came from Microsoft Copilot Auto. Across the twelve scored prompts it achieved 96.3%, compared with 78.8% for Google Gemini 3.1 Pro. In the ten-prompt controlled round, the comparison was closer: 98.0% versus 89.7%. The difference widened substantially in the two open-web stretch tasks, where source auditing, citation discipline and task execution became central.

The result should not be reduced to a product preference. The more useful outcome is a set of observable reasoning standards.

Good environmental AI reasoning:

- traces important claims to the source that actually supports them;
- distinguishes primary authority evidence from scientific mechanism evidence;
- respects spatial scale;
- preserves modal and conditional language;
- separates administrative status from measured ecological harm;
- distinguishes direct effects from visitor-mediated effects;
- treats uncertainty as part of the answer;
- and refuses to invent precision where the evidence is incomplete.

For the project itself, the practical implication is equally clear: future protected-landscape work should be permission-first, evidence-led and communication-aware. For AI evaluation, the case demonstrates that the hardest failures often occur not when a model lacks facts, but when it has many relevant facts and combines them too confidently.

Appendix A — Prompt bank

- 1 Geological explanation: eolianites, ooids, fossil dunes and current processes.
- 2 "Dry-laid means harmless?": known, plausible and site-specific evidence.
- 3 Analogue evidence: what the comparison site can and cannot establish at this case site.
- 4 Geoheritage impact pathways ranked by evidential confidence.
- 5 Biodiversity without invention at the exact footprint.
- 6 Protection layers: park management, EU habitat network, international geopark recognition, geosite and coastal public domain.
- 7 Evidence-led documentation of removal and subsequent site condition.
- 8 Low-impact research design for visitor behaviour and online promotion.
- 9 Balanced critique of the project's stated environmental ethic and the installation.
- 10 Environmental strengths and limits of moving from stone to book/browser-based experience.
- 11 Open-web audit of current public copy after the authority correspondence.
- 12 Permission-first environmental due-diligence checklist for future outdoor interventions.

Appendix B — Principal sources

Institutional names, reference codes and direct URLs have been withheld or generalised throughout this appendix to protect the identity of the underlying project. The evidence types, dates and evidential function of each source are preserved.

S0 — Written communication from the relevant protected-area authority to the project, dated September 2026. Primary case evidence controlling the authority's statements on authorisation, public-land/georesource context, visitor promotion, removal, possible future sanctioning process and geopark/park collaboration.

S1A — Archived third-party labyrinth-directory record, checked August 2026. Supports the dated directory wording, size/date fields and historical access/promotion record.

S1B — Current project public web pages, accessed 27 September 2026. Supports the project's present description of the original installation and current public communications.

S2 — Regional government geological trail guide covering the fossil dune system. Supports Late Pleistocene fossil dunes, ooids, oolitic dune formation, cross-bedding and present coastal/alluvial processes.

S3 — National geological-heritage inventory record for the site. Supports geological-interest inventory status and principal sedimentological interest.

S4 — Regional EU habitat-network record for the wider protected area. Supports wider habitat/species context; not exact-footprint biodiversity proof.

S5 — Regional decree establishing the protected-area management framework. Supports the park's management framework.

S6 — International geopark recognition record. Supports international geopark context only; not project endorsement.

S7 — National coastal law, relevant article on uses of state-owned coastal public domain land. Supports the general legal framework for such uses.

S8 — Sanromualdo-Collado et al. (2023), *Science of the Total Environment* 894, 164934. DOI: <https://doi.org/10.1016/j.scitotenv.2023.164934>. External analogue on stone wind shelters, airflow, sediment transport and vegetation in a different active dune system on the Spanish coast.

S9 — Rocha et al. (2020), *Human–Wildlife Interactions* 14(1), Article 17. DOI: <https://doi.org/10.26077/secn-2a27>. General mechanism/context on stone displacement and rock-associated microhabitats.

Appendix C — Audit trail

The project archive retains:

- the source index and ground-truth matrix;
- locked prompt benchmarks and run packets;
- unedited raw responses for both models;
- per-prompt model metadata;
- blind scoring sheets with deduction reasons;
- post-reveal identity mappings;
- the current public-copy audit baseline for Prompt 11;
- and the final results synthesis.

Local and cloud copies were mirrored and SHA-256 checked during the evaluation workflow. Unredacted source records, including the withheld institutional identifiers in Appendix B, are held separately and can be produced on request under confidentiality.